

A top-down view of a laboratory workspace. A scientist wearing a white lab coat and white gloves is working with a multi-well plate. The plate contains various colored liquids and powders. The scientist is using a pipette to transfer liquid from one well to another. The background shows shelves with various laboratory bottles and containers.

Microanalysis in Pharmaceutical Product Development and Troubleshooting

Fast, non-destructive problem-solving in drug development

Through the conversion of visual abnormalities into chemical insights, this application note investigates how spatially resolved infrared microanalysis deepens pharmaceutical problem-solving. Beyond simple identification verification, it illustrates how FTIR microscopy closes the gap between observation and action by facilitating data-driven, direct decision-making in situations involving contamination, formulation flaws, or unforeseen variability. Tablets, biopharmaceutical solutions, and excipient mapping are used as case studies to show how focused spectrum analysis is a useful tool in quality control and research and development settings since it not only identifies a material but also explains its presence.

Keywords or phrases:

Pharmaceutical products, FTIR microscopy, quality control, troubleshooting, particles and contaminants

INTRODUCTION

The demand on the quality of pharmaceutical products is very high. Raw and packaging materials as well as intermediate and final products must contain the correct components and need to be free from contaminations. In quality control, analytical methods like Mid- and Near-IR spectroscopy and others allow to verify an overall correct identity and composition of the sample.

However, contaminants like particles in liquid formulations or inclusions in tablets are often extremely small and cannot be detected selectively by such macroscopic methods. Therefore, further analytical methods are needed to determine the accurate chemical nature of the particle or inclusion – valuable information that allows to find the source of contamination.

This application note demonstrates how FTIR microanalysis using the Bruker LUMOS II supports both troubleshooting and pharmaceutical development, offering fast, automated, and spatially resolved insights across a variety of sample types.

IDENTIFICATION OF AN INCLUSION INSIDE A PHARMACEUTICAL TABLET

Pharmaceutical tablets have to be free from inclusions. If an inclusion is detected e.g. by visual inspection its chemical nature has to be determined to raise the chances finding its origin. Figure 1 shows a white tablet with a yellow spot inside.



Figure 1. Pharmaceutical tablet with contamination (yellow).

To discriminate the inclusion from the general matrix of the tablet a series of positions was measured on the yellow spot and next to it. Using the fully automated ATR (Attenuated Total Reflectance) mode of the LUMOS II the region of interest was analysed without any sample preparation; the tablet just had to be fixed in a micro vice on the microscopic stage. Guided by the software microscopic images of the region of interest were taken and measurement positions defined before the measurement was performed by the LUMOS II automatically. An area of $10 \times 10 \mu\text{m}$ per spot was measured with the spectrum acquisition taking 10 seconds, respectively. Figure 2 shows the visual microscopic image of the inclusion inside the tablet together with the measurement positions. The spectra are coloured according to their position.

It is obvious that the spectra measured on the inclusion (blue) have a much different signature than the ones on the general matrix of the tablet (red).

Result of IR-microscopic measurement

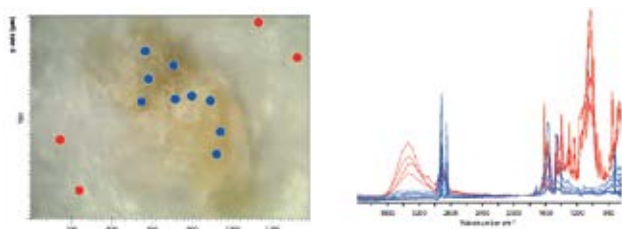


Figure 2. IR-microscopic mapping on pharmaceutical tablet with unwanted yellow inclusion. The spectra on the inclusion (blue) show a different signature compared to those from the tablet matrix (red)

The identification of the inclusion as magnesium stearate was achieved within a few seconds by searching the comprehensive spectral library ATR-COMplete which contains > 26.000 spectra of various substances. Figure 3 shows the high similarity of the library spectrum with the measured spectrum.

Similarity library spectrum / measured spectrum

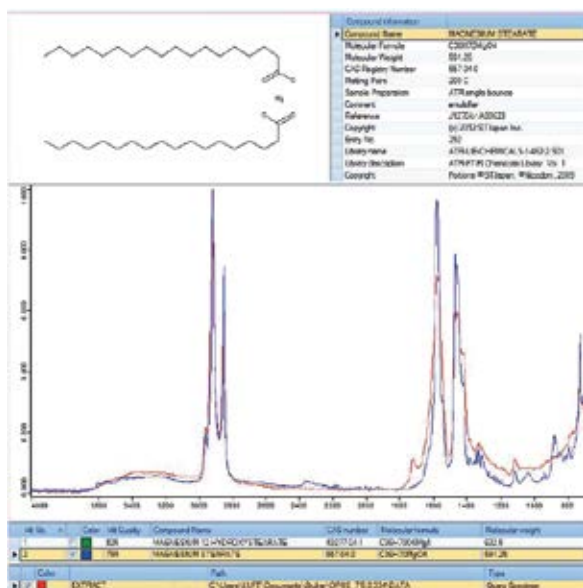


Figure 3. Clear identification of the inclusion as magnesium stearate by library search

DETERMINATION OF THE CHEMICAL IDENTITY OF A PARTICLE

Liquid sterile formulations must be free from all visible and even smaller particles. Unwanted particles might originate from the manufacturing equipment, the personnel or the used packaging material. Also unwanted precipitation of the active ingredient and excipient might result in particles.

Hence, the chemical composition of potential particles covers a wide range including plastic and rubber fragments, cellulose fibres, glass, metal or biological material like protein. For further analysis, particles are isolated from the liquid formulation using a gold filter (Figure 4) which can be used directly as substrate for the IR-microscopic measurement.



Figure 4. Gold filter for the isolation of particles from liquid formulations.

Figure 5 shows the microscopic image of several fibres that were filtered from the liquid formulation of a protein drug.

The spectra were measured without further sample preparation directly on the fiber bundle and on the filter substrate next to it with the LUMOS II by applying the ATR-technique. Whereas the spectra on the filter are typical for the protein drug (amide I at 1640 cm⁻¹ and amide II 1550 cm⁻¹; not shown) the fiber has a strong double peak around 1200 cm⁻¹.

Microscopic image and spectrum of particle

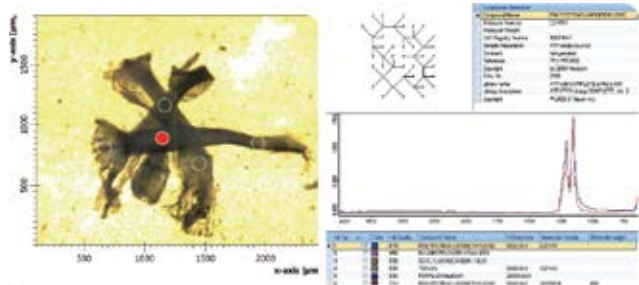


Figure 5. Left: Microscopic image of a bundle of fibres on a gold filter separated from a liquid biopharmaceutical. Right: Identification of the spectrum on the indicated measurement position as Poly(tetrafluoroethylene) (Teflon) by library search tablet covering an area of 500x325 µm. The spatial resolution of 25x25 µm (indicated by red rectangles) is defined by the operator using an automated physical aperture which is integrated in the LUMOS II.

ANALYSIS OF THE DISTRIBUTION OF API AND EXCIPIENTS IN A PHARMACEUTICAL TABLET

The quality of a pharmaceutical tablet is not only defined by its components but also by their distribution inside the tablet. As an example, a protective layer around the tablet as well as the homogeneity of the active pharmaceutical ingredient (API) distribution defines where and at which speed the API is released inside the body. IR-microscopic mapping measurements create chemical images of the API and the excipients distribution.

In this example, a fully automated ATR-mapping on the cross-section of a pharmaceutical tablet of the anti-inflammatory drug ibuprofen was performed using the LUMOS II. An area of 500x325 µm was measured with a spatial resolution of 25x25 µm (Figure 6).

Apart from the active pharmaceutical ingredient ibuprofen the excipients lactose, microcrystalline cellulose and sodium dodecyl sulphate are present in the tablet. To determine their distribution inside the tablet a linear combination of previously measured pure substance spectra was

performed to explain the composition of each individual mapping spectrum.

The calculation was done fully automatic using the OPUS spectroscopic software.

Rectangular mapping grid

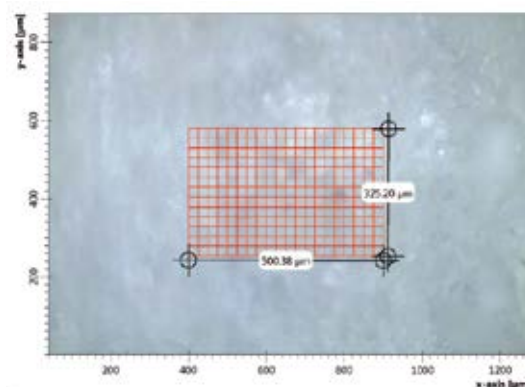


Figure 6. Rectangular map on the cross section of a pharmaceutical tablet covering an area of 500x325 µm. The spatial resolution of 25x25 µm (indicated by red rectangles) is defined by the operator using an automated physical aperture which is integrated in the LUMOS II.

The chemical images in Figure 7 visualize the composition of the tablet at each measurement spot using a colour scheme ranging from black (no contribution of component) to bright orange (high contribution of component).

Combination of the chemical images shows which part of the tablet is dominated by which component (Figure 8).

Chemical images

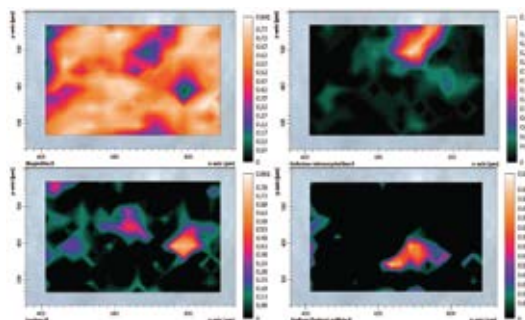


Figure 7. Chemical images of the distribution of different compounds in a pharmaceutical tablet: ibuprofen (upper left), microcrystalline cellulose (upper right), lactose (lower left) and sodium dodecyl sulphate (lower right).

CONCLUSION

IR-microanalysis is an extremely powerful and valuable technique to effectively track the source of product contaminations found in routine quality control. Furthermore, the spatially resolved analysis of pharmaceutical products like tablets or lyophilizates provides insight in their composition and homogeneity. This knowledge helps pharmaceutical research and development to their optimize products.

Using the compact FTIR microscope LUMOS II even inexperienced users are able to perform an IR-microscopic analysis with minimal training. It's full automation and intuitive software interface result in a user comfort that will save valuable time.

Combination of the chemical images

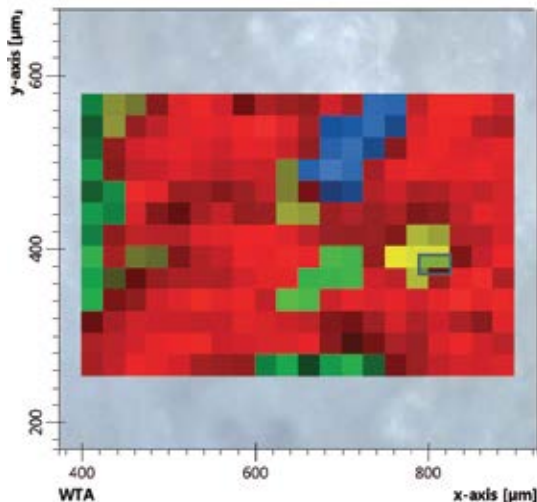


Figure 8. Combined chemical images of the distribution of different compounds in a pharmaceutical tablet: ibuprofen (red), microcrystalline cellulose (blue), lactose (yellow) and sodium dodecyl sulphate (green).



LUMOS II FTIR microscope